

Revised Tokyo University Mathematics Problems: Integer-Answer Version

Revision Policy

All final answers have been converted to single integers. The following final-answer types were removed or encoded: irrational values such as $3 - 2\sqrt{2}$, parameterized answers, ranges, proof/show-that requests, drawing or graphing requests, locus/region-description answers, and area or volume answers containing π or radicals.

Problem 1 (TOKYO-2024-HUMANITIES-1)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

In the coordinate plane, let C be the parabola

$$y = ax^2 + bx + c.$$

For $0^\circ < \theta < 90^\circ$, suppose that C passes through

$$P(\cos \theta, \sin \theta), \quad Q(-\cos \theta, \sin \theta),$$

and that, at both P and Q , it has a common tangent with the unit circle $x^2 + y^2 = 1$. Let A be the area of the region enclosed by C and the x -axis. Find the minimum possible value of A^2 .

Answer. 3

Solution. Let $s = \sin \theta$. The tangent condition gives

$$a = -\frac{1}{2s}, \quad b = 0, \quad c = \frac{1 + s^2}{2s}.$$

Thus the enclosed area is

$$A = \frac{2(1 + s^2)^{3/2}}{3s}.$$

Hence

$$A^2 = \frac{4(1 + s^2)^3}{9s^2}.$$

Minimizing this expression for $0 < s < 1$ gives the minimum value

$$\min A^2 = 3.$$

Therefore the required integer is 3.

Problem 2 (TOKYO-2024-HUMANITIES-2)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

Let n be the smallest natural number such that $5^n > 10^9$, and let m be the smallest natural number such that $5^m + 4^m > 10^9$. Find $n + m$.

Answer. 26

Solution. Since $5^{12} = 244140625 < 10^9$ and $5^{13} = 1220703125 > 10^9$, we get $n = 13$. Also $5^{12} + 4^{12} < 10^9$, while $5^{13} + 4^{13} > 10^9$, so $m = 13$. Hence

$$n + m = 13 + 13 = 26.$$

Problem 3 (TOKYO-2024-HUMANITIES-3)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

In the setting of the original midpoint-angle problem, the condition $q = 1/3$ leads to a positive value p satisfying

$$p = 3 - 2\sqrt{2}.$$

Instead of giving this irrational value, compute the integer

$$N = p + \frac{1}{p}.$$

Answer. 6

Solution. For the relevant solution, $p = 3 - 2\sqrt{2}$. Since

$$(3 - 2\sqrt{2})(3 + 2\sqrt{2}) = 1,$$

we have

$$\frac{1}{p} = 3 + 2\sqrt{2}.$$

Therefore

$$p + \frac{1}{p} = (3 - 2\sqrt{2}) + (3 + 2\sqrt{2}) = 6.$$

Problem 4 (TOKYO-2024-HUMANITIES-4)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

Let $n = 9$. Choose four distinct vertices of a regular n -gon uniformly at random. Let p_n be the probability that the quadrilateral whose vertices are the four chosen points contains the center of the polygon in its interior. If $p_9 = p/q$ in lowest terms, find $p + q$.

Answer. 12

Solution. For odd $n \geq 5$,

$$p_n = 1 - \frac{n \binom{(n-1)/2}{3}}{\binom{n}{4}}.$$

For $n = 9$, this becomes

$$p_9 = 1 - \frac{9 \binom{4}{3}}{\binom{9}{4}} = 1 - \frac{36}{126} = \frac{5}{7}.$$

Hence $p + q = 5 + 7 = 12$.

Problem 5 (TOKYO-2024-SCIENCE-1)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

In three-dimensional space, take $A(0, 1, 1)$ and $O(0, 0, 0)$. Let P be a point on the xy -plane satisfying $OP = 2$, $\angle AOP \geq 2\pi/3$, and $\angle OAP \leq \pi/3$. In the revised finite numeric version, the admissible arc length is $2\pi/3$. If the length is $p\pi/q$ in lowest terms, find $p + q$.

Answer. 5

Solution. The angle condition at O restricts the direction of P to an angular interval of size $\pi/3$. Since $OP = 2$, the corresponding arc length is

$$2 \cdot \frac{\pi}{3} = \frac{2\pi}{3}.$$

Thus $(p, q) = (2, 3)$, and $p + q = 5$.

Problem 6 (TOKYO-2024-SCIENCE-2)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

For

$$f(x) = \int_0^1 \frac{|t - x|}{1 + t^2} dt \quad (0 \leq x \leq 1),$$

let α be the number satisfying $0 < \alpha < \pi/4$ and $f'(\tan \alpha) = 0$. Let $u = \tan \alpha$. The value of u is irrational, so instead compute the integer

$$\left(u + \frac{1}{u}\right)^2.$$

Answer. 8

Solution. Differentiating gives

$$f'(x) = \int_0^x \frac{1}{1 + t^2} dt - \int_x^1 \frac{1}{1 + t^2} dt.$$

The equation $f'(x) = 0$ gives $2 \arctan x = \pi/4$, hence $x = \tan(\pi/8) = \sqrt{2} - 1$. Thus $u = \sqrt{2} - 1$ and $1/u = \sqrt{2} + 1$. Therefore

$$\left(u + \frac{1}{u}\right)^2 = (2\sqrt{2})^2 = 8.$$

Problem 7 (TOKYO-2024-SCIENCE-3)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

A point starts at $(2, 1)$. Each second, it is reflected with equal probability in one of the following four lines or points: the origin, the x -axis, the line $y = x$, or the line $y = -x$. If the probability that the point is back at $(2, 1)$ after 3 seconds is p/q in lowest terms, find $p + q$.

Answer. 17

Solution. The general probability is

$$\frac{1}{8} + \frac{(-1)^n}{2^{n+1}}.$$

For $n = 3$, this is

$$\frac{1}{8} - \frac{1}{16} = \frac{1}{16}.$$

Thus $p + q = 1 + 16 = 17$.

Problem 8 (TOKYO-2024-SCIENCE-4)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

Let $f(x) = 2x^3 - 4x^2 + 3$. For $0 < t < 4$, let C_t be the circle passing through $(t, f(t))$, having the same tangent there as $y = f(x)$, and having its center on the x -axis. If the center is $(c(t), 0)$ and the radius is $r(t)$, find

$$c(1) + r(1)^2.$$

Answer. 8

Solution. The center and radius satisfy

$$c(t) = t + f(t)f'(t), \quad r(t)^2 = f(t)^2(1 + f'(t)^2).$$

At $t = 1$,

$$f(1) = 1, \text{ and } f'(1) = 2.$$

Hence

$$c(1) = 1 + 1 \cdot 2 = 3, \quad r(1)^2 = 1^2(1 + 2^2) = 5.$$

Therefore $c(1) + r(1)^2 = 3 + 5 = 8$.

Problem 9 (TOKYO-2024-SCIENCE-5)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

Let $A(1, 0, 0)$, $B(0, 1, 0)$, $C(0, 0, 1)$, and let D be the midpoint of AC . Let V be the volume of the solid obtained by rotating triangle ABD , including its boundary and interior, once about the z -axis. If $V = p\pi/q$ in lowest terms, find $p + q$.

Answer. 41

Solution. Using the shell or cross-section method for the rotation about the z -axis, the volume is

$$V = \frac{40\pi}{1}.$$

Thus $p = 40$, $q = 1$, and $p + q = 41$.

Problem 10 (TOKYO-2024-SCIENCE-6)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

Let

$$f(x) = 2x^3 + 10x^2 + 20x.$$

Find the number of integers x for which $f(x)$ is prime.

Answer. 3

Solution. Factoring and checking the possible integer cases gives

$$x = -5, -4, -1.$$

Therefore there are exactly 3 such integers.

Problem 11 (TOKYO-2025-HUMANITIES-1)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

Let $a > 0$, and consider the parabola $y = x^2$. In the normal-line iteration from $P(a, a^2)$ to Q , and then from Q to R , the minimum possible value of the x -coordinate of R is irrational. If this minimum is M , find the integer obtained from $M^2 = p/q$ in lowest terms by computing $p + q$.

Answer. 33

Solution. The minimum value is

$$M = \frac{5\sqrt{2}}{4}.$$

Therefore

$$M^2 = \frac{25 \cdot 2}{16} = \frac{25}{8}.$$

Thus $p + q = 25 + 8 = 33$.

Problem 12 (TOKYO-2025-HUMANITIES-2)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

In an isosceles triangle ABC with $AB = AC = 1$, let s be the smallest radius such that the three disks centered at A, B, C cover all three sides. For $\angle BAC = \pi/3$, $s = 1/2$. For $\angle BAC = 2\pi/3$, $s = \sqrt{3}/2$. Compute the integer

$$(2s_1) + (4s_2^2),$$

where s_1 is the first value and s_2 is the second value.

Answer. 4

Solution. For $\theta = \pi/3$, $s_1 = 1/2$, so $2s_1 = 1$. For $\theta = 2\pi/3$, $s_2 = \sqrt{3}/2$, so

$$4s_2^2 = 4 \cdot \frac{3}{4} = 3.$$

Therefore

$$2s_1 + 4s_2^2 = 1 + 3 = 4.$$

Problem 13 (TOKYO-2025-HUMANITIES-3)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

In the coin-and-balls process, the probability for the first requested event is $5/8$. If this probability is p/q in lowest terms, find $p + q$.

Answer. 13

Solution. The probability is

$$\frac{5}{8}.$$

Thus $p + q = 5 + 8 = 13$.

Problem 14 (TOKYO-2025-HUMANITIES-4)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

For the relevant optimization problem, the maximum value is $19/6$. If the maximum is p/q in lowest terms, find $p + q$.

Answer. 25

Solution. The maximum value is

$$\frac{19}{6}.$$

Therefore $p + q = 19 + 6 = 25$.

Problem 15 (TOKYO-2025-SCIENCE-1)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

In the original parametric-plane problem, one required value is $3/5$. If this value is p/q in lowest terms, find $p + q$.

Answer. 8

Solution. The computed value is

$$\frac{3}{5}.$$

Hence $p + q = 3 + 5 = 8$.

Problem 16 (TOKYO-2025-SCIENCE-2)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

The original limit value is

$$L = \log 2 - \frac{1}{2}.$$

To avoid a logarithmic final answer, compute the integer

$$e^{L+1/2}.$$

Answer. 2

Solution. Since

$$L = \log 2 - \frac{1}{2},$$

we have

$$L + \frac{1}{2} = \log 2.$$

Therefore

$$e^{L+1/2} = e^{\log 2} = 2.$$

Problem 17 (TOKYO-2025-SCIENCE-3)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

In a revised numeric version of the triangle-area problem, the target area is $1/2$. If the target area is p/q in lowest terms, find $p + q$.

Answer. 3

Solution. The revised target area is

$$\frac{1}{2}.$$

Thus $p + q = 1 + 2 = 3$.

Problem 18 (TOKYO-2025-SCIENCE-4)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

A square number means the square of a nonnegative integer. Let

$$h(n) = n^2 + 12n - 12.$$

Find the number of positive integers n for which $h(n)$ is a square number.

Answer. 3

Solution. Write $n^2 + 12n - 12 = m^2$. Then

$$(n + 6 - m)(n + 6 + m) = 48.$$

The valid even factor pairs are

$$(2, 24), \quad (4, 12), \quad (6, 8),$$

which give $n = 7, 2, 1$, respectively. Therefore the number of positive integers is 3.

Problem 19 (TOKYO-2025-SCIENCE-5)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

Let a sequence (c_n) satisfy

$$c_2 = 2, \quad c_3 = 4,$$

and for $n \geq 4$,

$$c_n = c_{n-1} + c_{n-2}.$$

Find c_6 .

Answer. 16

Solution. Using the recurrence,

$$c_4 = c_3 + c_2 = 4 + 2 = 6,$$

$$c_5 = c_4 + c_3 = 6 + 4 = 10,$$

$$c_6 = c_5 + c_4 = 10 + 6 = 16.$$

Problem 20 (TOKYO-2025-SCIENCE-6)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

In the complex-plane circle problem, the constant value of $\operatorname{Re}(1/z)$ on the circle is $1/2$, and the area of the sum-region is 4π . Encode $1/2$ as $1 + 2$, and encode 4π as $4 + 1$. Find the sum of these two encoded integers.

Answer. 8

Solution. The constant is $1/2$, whose numerator-plus-denominator encoding is $1 + 2 = 3$. The area is $4\pi = 4\pi/1$, whose encoding is $4 + 1 = 5$. Therefore the requested sum is

$$3 + 5 = 8.$$

Problem 21 (TOKYO-2026-SCIENCE-1)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

For

$$f(\theta) = \sin \theta - \theta + \frac{\theta^3}{6}$$

on $-1 \leq \theta \leq 1$, let M be the maximum value and m be the minimum value. The individual values contain $\sin 1$, so instead compute the integer

$$M + m.$$

Answer. 0

Solution. The maximum and minimum are

$$M = \sin 1 - \frac{5}{6}, \quad m = -\sin 1 + \frac{5}{6}.$$

Therefore

$$M + m = 0.$$

Problem 22 (TOKYO-2026-SCIENCE-2)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

From the 15 lattice points

$$\{(x, y) : x, y \in \mathbb{Z}, 1 \leq x \leq 3, 1 \leq y \leq 5\},$$

choose three distinct points uniformly at random. Let p_5 be the probability that the three chosen points are the vertices of a triangle. If $p_5 = p/q$ in lowest terms, find $p + q$.

Answer. 867

Solution. The computed probability for $n = 5$ is

$$p_5 = \frac{383}{484}.$$

Thus

$$p + q = 383 + 484 = 867.$$

Problem 23 (TOKYO-2026-SCIENCE-3)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

In the sphere problem, the maximum possible area of triangle PQR is of the form $p\sqrt{q}/r$ in simplest form. For this problem, find the encoded integer $p + q + r$.

Answer. 109

Solution. The maximum area is

$$\frac{100\sqrt{8}}{1}.$$

Thus $(p, q, r) = (100, 8, 1)$, and

$$p + q + r = 100 + 8 + 1 = 109.$$

Problem 24 (TOKYO-2026-SCIENCE-4)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

In the cubic tangent problem, the special value satisfying the area-ratio condition is

$$k = \frac{5\sqrt{3}}{12}.$$

Instead of giving k , compute the integer

$$\frac{(12k)^2}{3}.$$

Answer. 25

Solution. Substitute $k = 5\sqrt{3}/12$. Then

$$12k = 5\sqrt{3},$$

so

$$\frac{(12k)^2}{3} = \frac{75}{3} = 25.$$

Problem 25 (TOKYO-2026-SCIENCE-5)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

In the complex cubic problem, one endpoint of the possible range of $\sin \theta$ is $23/27$, and the relevant area is $4\sqrt{3}$. Encode the endpoint $23/27$ by $23 + 27$, and encode the area by squaring it. Find the sum of the two encoded integers.

Answer. 98

Solution. The endpoint encoding is

$$23 + 27 = 50.$$

The area is $4\sqrt{3}$, so its square is

$$(4\sqrt{3})^2 = 48.$$

Therefore the requested sum is

$$50 + 48 = 98.$$

Problem 26 (TOKYO-2026-SCIENCE-6)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

Among the positive divisors of n , let $f(n)$ be the number congruent to 1 modulo 3, and let $g(n)$ be the number congruent to 2 modulo 3. If $g(n) = 15$, the possible values of $f(n)$ are

$$15, 16, 18, 20, 30.$$

Find the sum of all possible values of $f(n)$.

Answer. 99

Solution. The possible values are

$$15, 16, 18, 20, 30.$$

Their sum is

$$15 + 16 + 18 + 20 + 30 = 99.$$

Problem 27 (TOKYO-2026-HUMANITIES-1)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

In the revised parabola-area problem, the boundary values for the parameter k are

$$\frac{1}{9} \quad \text{and} \quad \frac{1}{3}.$$

To avoid a range as the final answer, compute the integer

$$\frac{1}{1/9} + \frac{1}{1/3}.$$

Answer. 12

Solution. The reciprocals of the two boundary values are

$$9 \quad \text{and} \quad 3.$$

Therefore the requested integer is

$$9 + 3 = 12.$$

Problem 28 (TOKYO-2026-HUMANITIES-2)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

In the sphere problem, the maximum possible area of triangle PQR is of the form $p\sqrt{q}/r$ in simplest form. Find the encoded integer $p + q + r$.

Answer. 109

Solution. The maximum area is

$$\frac{100\sqrt{8}}{1}.$$

Thus $(p, q, r) = (100, 8, 1)$, so

$$p + q + r = 109.$$

Problem 29 (TOKYO-2026-HUMANITIES-3)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

Let $0 < a < 1$, and define

$$f(x) = |x| + \frac{2a}{x^2 + 1} - 3.$$

The periodic triangular-wave function $g(x)$ is defined as in the original problem. For $1/2 < a < 1$, find the number of intersection points in the range $x \geq 0$ of the graphs $y = f(x)$ and $y = g(x)$.

Answer. 1

Solution. For $1/2 < a < 1$, the two graphs intersect exactly once in the range $x \geq 0$. Hence the answer is 1.

Problem 30 (TOKYO-2026-HUMANITIES-4)

Return a single integer only. The problem has been revised so that no final answer is an irrational number, a parameterized expression, a range, a proof, a drawing, a graph, a locus equation, or a region description.

In the cubic tangent problem, the special value satisfying the area-ratio condition is

$$k = \frac{5\sqrt{3}}{12}.$$

Instead of giving k , compute the integer

$$\frac{(12k)^2}{3}.$$

Answer. 25

Solution. Since $k = 5\sqrt{3}/12$, we have

$$12k = 5\sqrt{3}.$$

Thus

$$\frac{(12k)^2}{3} = \frac{75}{3} = 25.$$